

AI-POWERED CAD ANALYSIS REPORT

Generated: 12 Aug 2026

Geometry to Process to Cost · Fully AI-reasoned

46

Casting Bracket

133 × 120 × 125 mm · 320.0 cm³ · AI 0.864 kg / Steel 2.512 kg

Manufacturability: 46/100 · Confidence: High

§1 — Geometry & Part Summary

Bounding Box	132.6 × 120.0 × 125.0 mm	Surface Area	628.3 cm ²
Volume	320.01 cm ³	Weight (AI)	0.864 kg
Weight (Steel)	2.512 kg	Weight (Plastic)	0.336 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
Holes	10	High	10 through/blind holes, radii 2-10mm requiring core pins or post-cast drilling
Threaded feature	1	Medium	Threaded boss detected — likely requires tapping or cast-in insert
Free-form surfaces	54	High	54 B-spline faces forming organic/blended bracket geometry, favouring casting over machining
Undercuts	21	High	21 undercut faces requiring side-cores/loose pieces or re-orientation of parting line
Non-uniform wall section	1	High	Wall thickness ranges 1.54-30.21 mm (Å=8.83 mm) — mix of thin bosses typical of a structural sand/gravity casting

§3 — Material Analysis

AI-suggested

Primary Material:

Ductile (SG) Iron GJS500 (55% confidence)

No aluminium signal (min wall 1.54mm is local, but mean 17.6mm/max 30.2mm is far too thick for HPDC aluminium). The part is a structural bracket with heavy, non-uniform sections and 21 undercuts consistent with a sand-cast ductile iron bracket used in chassis/powertrain mounting applications. Following the guideline to prefer steel/iron over aluminium for thick structural brackets absent an explicit lightweighting signal.

Alternatives:

LM25 Aluminium Casting Alloy (25%) · Grey Iron GJL350 (15%) · Austempered Ductile Iron (5%)

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Sand casting (automated green-sand line)	casting	65%	0.2410	Thick, non-uniform sections and undercuts favour sand casting; high volume (200k/yr) supports an automated match-plate/DISA-line sand process with cast iron.
Gravity die casting	casting	20%	0.1850	Feasible if aluminium is confirmed; would reduce cycle time but tooling life shorter for iron.
Cast + machine finishing	cast_and_machine	15%	0.2760	Post-cast machining required for 10 holes, threads and mating faces.

§5 — Manufacturability Risks (Score: 46/100)

Severity	Feature / Area	Description	Recommended Action
High	Undercuts (21 faces)	Numerous undercut faces will require side-cores, loose pieces or a redesigned parting line, increasing pattern/tooling complexity and scrap risk.	Re-orient part or split into cast + machined sub-features; consolidate undercuts to a single core where possible.
Medium	Non-uniform wall thickness (1.54-30.21mm)	Large variation promotes shrinkage porosity at thick bosses and cold-shut risk at thin ribs.	Add fillets/tapered transitions and risers at heavy sections; core out bosses >20mm.
Medium	Threaded feature	Cast-in threads are not achievable reliably; requires tapping or heli-coil insert operation post-cast.	Cast a pilot hole and tap after machining.
Low	Free-form B-spline surfaces (54)	Organic surfaces increase pattern-making cost and inspection complexity.	Simplify blends where not functionally required.

§6 — DFM Issues (casting)

Severity	Area	Description	Impact	Fix
High	Undercuts / parting line	21 undercut faces will require side-cores or loose pattern pieces, raising pattern cost and cycle time.	Increases pattern tooling cost by 20-40% and adds core-setting labour per shot.	Redesign features to draw in single pull direction or accept a segmented core; consolidate undercuts into fewer cores.
High	Wall thickness variation (1.54-30mm)	Thin ribs adjacent to thick bosses (up to 30mm) create differential cooling rates, causing shrink porosity and hot tears at junctions.	Scrap/rework rate rises, X-ray/inspection cost increases for structural certification.	Add fillets and gradual wall transitions; place risers directly over heavy bosses; core out solid sections >20mm to even wall thickness (~8-12mm).
Medium	Threaded feature	Threads cannot be cast net-shape reliably in sand casting.	Requires secondary tapping operation, adding machining time and setup.	Cast a pilot/clearance hole sized for tapping; specify thread class post-machining.
Medium	10 small holes (2-10mm radius)	Small-diameter holes are difficult to cast cleanly and may need core pins prone to shift/breakage in high-volume automated moulding.	Core shift causes hole mislocation, driving scrap and rework.	Cast larger pilot holes and finish-machine to size, or use robust core prints with locating features.
Low	Free-form B-spline surfaces	Complex blended surfaces increase pattern equipment cost and inspection time.	Pattern-making and CMM inspection cost increase moderately.	Simplify non-functional blends to standard fillet radii.

§7 — Cost Range & Suggested Inputs

OPTIMISTIC

£15.50

MOST LIKELY

£22.00

CONSERVATIVE

£32.00

Net Weight	2.288 kg	Material	mat-gjs500
Recommended Process	casting	Cycle Time	0.2410 hr/part
Setup Time	0.750 hr	Operations	3 ops
Operations Detail	Sand mould & pour (ferrous) (mach-vmc3, 0.2410 hr) Fettling/deburr (mach-drill, 0.0500 hr) Drill/tap/finish machining (mach-vmc3, 0.2760 hr)		

Process-Specific Parameters

Casting Subtype	sand
Die/Mould Cost	£14,500
Die Life	8,000 shots
Cavities	2
Yield	65.0%
Flash Weight	0.000 kg
Yield	0.0%
Die Cost	£0
Strokes	0
Cavities	0
Wall Thickness	0 mm
Mould Cost	£0
Mould Life	0 shots
Projected Area	0.0 cm²

§8 — AI Analysis Explanation

This is a structurally complex bracket with thick, non-uniform wall sections (1.5-30mm), 21 undercuts and heavy free-form content (54 B-spline faces of 230 total), consistent with a sand or gravity cast component rather than thin-wall HPDC. Given the heavy, load-bearing geometry and absence of any explicit aluminium/lightweighting signal, ductile iron (GJS500) sand casting is recommended as the primary material/process, matching the user-forced 'casting' commodity. At 200,000 units/year an automated green-sand line with 2 cavities per mould and a ferrous cycle time of 0.241hr is assumed, using the deterministic parametric tooling cost of £14,500 for the pattern equipment. Post-cast machining (drilling 10 holes, tapping threads, fettling) adds the CNC bottom-up cycle time of 0.276hr per part. The 21 undercuts and non-uniform wall thickness are flagged as the top manufacturability risks requiring core/pattern redesign.

§9 — Analysis Limitations & Assumptions

1. Material (cast iron vs aluminium vs steel) could not be confirmed from geometry alone; inferred from wall thickness, mass, and structural bracket typology per guideline.
2. Cavity count and mould life are rule-of-thumb estimates for a 200k/yr automated sand line, not directly measurable from geometry.
3. Thread specification (size/class) not determinable from provided geometry data.
4. Cost range assumes UK foundry labour/overhead rates and standard ductile iron raw material pricing; actual quotes may vary $\pm 25\%$.

Stage-1 pre-selection: auto (0%) ·